

Cryptanalysis of Multi-Chaos-Based Image Encryption Scheme for IoT

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Abstract

A lightweight multiple map chaos-based image encryption algorithm (MM-CBIE) was proposed recently for securing images in IoT systems. It uses Henon and two-dimensional logistic chaotic transforms along with a key mixing method in a permutation–diffusion structure. The scheme claims resistance to statistical and differential attacks.

This paper analyzes MMCBIE and shows it is vulnerable to a chosen plaintext attack. The weakness comes from its deterministic and separable design. The attack uses constant-value images as inputs and compares the resulting ciphertexts. Due to weak pixel-level diffusion and repetitive behavior in the key mixing stage, identical input values produce similar output patterns. This allows recovery of plaintext through direct equality matching.

A key issue is channel dependency in the key mixing phase. Each color channel is derived from another using the same modular addition. By analyzing equality relations among ciphertexts from selected inputs, attackers can reconstruct plaintext pixels without needing the chaotic sequences or secret keys.

The attack requires at most 256 oracle queries and has linear complexity relative to image size. Experiments confirm full image recovery, proving MMCBIE is not secure against chosen plaintext attacks and lacks proper

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inter-pixel and inter-channel dependency.

Keywords:

Chaotic image encryption, Cryptanalysis, Chosen-plaintext attack, Equality matching attack, Weak diffusion, Channel dependency, Henon map, Logistic map, IoT security

1. Introduction

The increased use of IoT devices has led to a revolution in modern communication infrastructure, facilitating instant transfer of data between intelligent home devices, industrial machinery, telemedicine applications, and even surveillance technology applications [1]. This revolution is significant, as recent studies reveal that billions of such devices have been put into use, transmitting confidential information through network channels [2].

A considerable amount of information that will be produced and transmitted through these devices consists of digital images; they may include medical scans, biometric information, satellite images, and video streams. All of these types of digital information are highly sensitive and require high levels of confidentiality [3]. Particularly when used in the healthcare industry, there are strict regulatory requirements for the security of these images.

Since IoT devices have very limited resources in terms of processing power, storage, and battery life, cryptographic techniques like AES become highly computationally expensive when applied directly to high resolution images [4]. Similarly, Triple DES becomes impractical under the constraints of IoT bandwidth and latency issues [5]. The same applies for public key algorithms like RSA.

This challenge has resulted in an increasing number of studies on lightweight image-based cryptographic algorithms that can run effectively within IoT limitations but offer reasonable security [6]. Chaotic systems in parallel have been suggested as affordable building blocks for lightweight algorithms for securing green IoT applications [7], and grain-based stream ciphers with chaotic systems have been proposed as solutions to secure low-power image processing in IoT networks [8]. Another approach to securing images transmitted in IoT networks is watermarking and lightweight cryptography, which has been studied as a two-layer security mechanism [9].

Image encryption using chaos has been found to be the most effective approach to solve this problem [10]. Chaotic dynamical systems are con-

sidered suitable candidates for use in cryptography owing to their inherent characteristics such as sensitivity to initial conditions, ergodicity, mixing, and pseudorandomness, which directly correlate with confusion and diffusion in a cipher system [11]. The logistic map is one of the simplest chaos maps that has been extensively used for image encryption owing to its uniform distribution property and lower complexity [12]. Two-dimensional logistic map variants have been observed to possess better entropy and mixing capabilities than one-dimensional maps [13].

Hyperchaotic maps, which possess multiple positive Lyapunov exponents, have been proposed to further increase the complexity of the generated keystream [14]. The Hénon map, a canonical two-dimensional chaotic system, has been applied to image encryption through fractional Fourier transform frameworks to achieve strong pixel-level diffusion [15].

Parametric polynomial chaotic systems were presented to deal with the issue of performance deterioration and discontinuous parameters for many existing chaotic maps [16]. The techniques to create n -dimensional hyperchaotic systems with positive Lyapunov exponents have been investigated through the use of the Gershgorin criterion [17]. Chaotic Latin square-based substitution-boxes were designed to attain maximum algebraic nonlinearity in image encryption schemes [18].

Although chaotic systems have proven useful for encryption, it has been shown that statistical robustness does not necessarily guarantee security against cryptanalysis, especially with regard to chosen plaintexts [19]. In her pioneering research into symmetric ciphers derived from chaotic mappings in two dimensions, Fridrich made great strides and yet fell prey to attacks by chosen plaintexts and known plaintexts capable of retrieving secret chaotic variables without exhaustive key search [10].

Analysis of the QCMDC-IEA cipher that integrates quantum chaotic mapping and DNA encoding revealed that pseudo-random number generators independent of the plaintext generate keys that lead to the same results. Complete decryption using low complexity is possible using this vulnerability. A dual compression/encryption method using chaos permutation and row/column transformation is thus proposed [20].

Cryptanalyzing the ICIC-DNA color image cryptosystem revealed that various DNA manipulations could be reduced to two-bit substitutions, and that substitution and permutation stages could be independently attacked using divide-and-conquer approaches [21]. IES-M-BD, a cryptosystem based on memristive chaotic maps and a bidirectional cyclic bit shifting and dif-

fusion mechanism at the DNA level, is an example of the move towards multiple component schemes, which are more resilient against attacks of this nature [22]. Another high-security color image encryption scheme, the CIES-DVEM scheme, relies on dynamic vector-level manipulations and a two-dimensional logistic modularity map [23]. In the domain of multimedia IoT security, compressed sensing techniques for encryption, together with linear feedback shift register scramblers, have been investigated [24].

Selectively encrypting light-weight pixels in multimedia IoT applications is found to lead to power savings and reduction in packet rate with end-to-end privacy [25]. CES-Blocks is an example of a blockchain-based medical image transmission technique which utilizes a chaotic encryption algorithm in IoT that can withstand differential attacks [26].

Real-time RGB image encryption for IoT applications based on enhanced sequences from logistic 1D map models and MQQT-driven machine to machine communication has been found to exhibit throughputs comparable to embedded processors [27]. The ETS-ZSC scheme, which is comprised of the Enhanced Thorp shuffle and Zig-zag scan convolution techniques, enables one-time key security with NPCR and UACI close to optimal.

A scheme for the chaos-based encryption of images through confusion, diffusion, and chaotic-map encryption keys is supported by tests with NPCR and UACI metrics for smart cities [28]. An important example of the multi-map approach to chaotic map encryption was the guan2005chaos method that combined the Arnold's cat map with the Chen's chaotic map but was later proven to be vulnerable to chosen-plaintext attacks [29]. The LSC-IES algorithm integrates two chaotic maps and the cosine transformation to mitigate any non-chaotic behavior [13].

The major flaw in most image encryption algorithms utilizing the chaotic nature is inadequate diffusion along with deterministic, key-independent transforms [19]. This means that there will still be some correlation between the plaintext and the ciphertext which will facilitate attack even without the knowledge of the key. Non-flat surface level pyramid interconnection models have been suggested for IoT devices as they can provide hierarchical data abstraction for multidimensional images [30].

The use of fat-tree topologies with switch buffer blocking techniques results in less head-of-line blocking problems and shallower buffers, resulting in reduced energy consumption in IoT switching fabric design [31]. In this light, the current study analyzes the Multiple Map Chaos based Image Encryption (MMCBIE) algorithm designed by Jain et al. [32].

MMCBIE has the following three consecutive steps: (i) Pixel-level XOR with a matrix obtained by applying Henon-map; (ii) Cyclic key mixing operation performed among colors; and (iii) Pixel-level XOR with 2D logistic map. Although very good statistical measures like NPCR, UACI, Shannon Entropy, and SSIM have been claimed, it lacks adequate diffusion capabilities.

These assertions are proven false in this paper under a chosen plaintext attack scenario. The core weakness lies in the basic design flaw whereby encryption is deterministic and separate, lacking an efficient coupling of pixels. Encryption of constant images results in predictable output that leads to a straightforward chosen plaintext attack by matching the plaintext values with the target ciphertext through equality. This is done by using constant probe images for all the different values of the pixel. A comparison is then done between the ciphertext generated and the target ciphertext per channel, exploiting the cyclic nature of key mixing and channel dependency. Matching plaintexts and ciphertexts will result in obtaining the original plaintext without decrypting chaotic sequences or secret keys.

Contributions

The contributions of this paper are as follows.

1. **Analysis of structural vulnerabilities.** We analyze the multi-step nature of MMCBIE and demonstrate that the scheme does not provide adequate diffusion and suffers from channel-wise deterministic dependencies. Such properties allow ciphertext patterns to stay consistent relative to specific inputs, thus facilitating plaintext retrieval through equality tests.
2. **Practical equality-based CPA.** We devise an efficient CPA that retrieves plaintext values using constant images as inputs to the encryption oracle. Subsequent comparison of generated ciphertexts against the target ciphertext allows for equality-based plaintext value retrieval. The attack requires no more than 256 oracle calls and can be completed in $\mathcal{O}(n^2)$ time relative to the total number of pixels.
3. **Proof of concept.** We implement the cryptographic scheme and carry out the attack on standard test image. It is demonstrated that the attacked plaintext image matches the original plaintext image precisely.
4. **Recommendations for future design.** We trace the source of vulnerability to poor diffusion and plaintext independency and recommend

implementing strong intra-pixel dependencies and plaintext-dependent initialization for protection against such attacks.

The remainder of the paper is organised as follows. Section 2 describes the MMCBIE algorithm. Section 3 details the proposed attack. Section 4 reports experimental results. Section 5 concludes the paper.

2. Description of the Target Encryption Scheme

This section provides a concise description of the Multiple Map Chaos Based Image Encryption scheme, MMCBIE, proposed by Jain et al. [32], which serves as the target of the cryptanalytic attack developed in subsequent sections.

2.1. Chaotic Maps

MMCBIE is based on two types of chaotic maps. Hénon chaotic map is used for the first transformation phase through confusing the pixel values using pixel-wise XOR process. The last diffusion stage uses the Two-Dimensional Logistic Chaotic Map (2D-LSCM) using the same XOR process. Key-mixing step is placed between the two phases deterministically.

2.1.1. Hénon Chaotic Map

The Hénon map [33] is a classical two-dimensional discrete-time chaotic system defined by the recurrence

$$x_{i+1} = 1 - a x_i^2 + y_i, \quad (1)$$

$$y_{i+1} = b x_i, \quad (2)$$

with $x_i, y_i \in \mathbb{R}$ as the state variables of iteration i , and a, b as system parameters. In case of standard values $a = 1.4$, $b = 0.3$ the map shows chaotic behaviour in its dynamics with strange attractor; for the purposes of MMCBIE scheme, these structural constants are kept fixed, and only the initial state (x_0, y_0) is considered secret. Chaos matrix is then obtained from the dynamical sequence $(x_i, y_i)_{i=1}^{MN}$ by means of the quantization method

$$H_{r,c} = \left[(|x_i| + |y_i|) \times 2^{16} \right] \bmod 256, \quad i = (r-1)N + c, \quad (3)$$

where indices (r, c) run over rows and columns of the image respectively.

2.1.2. Two-Dimensional Logistic Chaotic Map (2D-LSCM)

The 2D-LSCM employed by MMCBIE [13] is a coupled nonlinear recurrence given by

$$x_{i+1} = r(3y_i + 1)x_i(1 - x_i), \quad (4)$$

$$y_{i+1} = r(3x_{i+1} + 1)y_i(1 - y_i), \quad (5)$$

where $(x_i, y_i) \in (0, 1)^2$ and r is the chaotic control parameter. The authors recommend the range $r \in [1.11, 1.19]$ to ensure optimal chaotic behaviour [13]. The warm-up process consists of an iteration of 500 steps that precedes the collection of output values for transient effects removal. A chaos matrix $L \in \{0, \dots, 255\}^{M \times N}$ is built from the trajectory $(x_i, y_i)_{i=1}^{MN}$ using the same quantisation rule as Eq. (3):

$$L_{r,c} = \left\lfloor (|x_i| + |y_i|) \times 2^{16} \right\rfloor \bmod 256, \quad i = (r - 1)N + c. \quad (6)$$

The chaos matrices H and L depend only on the corresponding initial conditions and structural constants; neither of them relates to the plaintext image.

2.2. Key Structure

The master secret key K of MMCBIE is a 512-bit quantity partitioned into six subkeys as follows:

$$K = (x_H, y_H, MK, x_L, y_L, R), \quad (7)$$

where $x_H, y_H \in \mathbb{R}$ (each represented as a 64-bit floating-point value) are the initial conditions supplied to the Hénon map; MK is a 192-bit key-mixing sub-key partitioned into three byte arrays $MK_R, MK_G, MK_B \in \{0, \dots, 255\}^8$, one per RGB colour channel; and $x_L, y_L \in (0, 1)$ and $R \in [1.11, 1.19]$ (each 64-bit). These constitute the input parameters for the 2D-LSCM. The one requirement for all subkeys is that R lies within the optimal chaotic domain of the 2D-LSCM.

2.3. Encryption Process

Let $P \in \{0, \dots, 255\}^{M \times N \times 3}$ be a plain colour image of height M and width N . MMCBIE applies three successive transformation stages to produce the cipher image C .

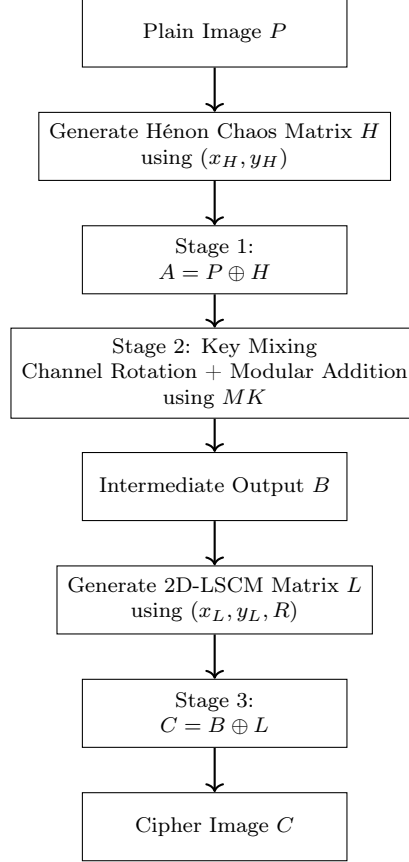


Figure 1: Encryption flow of MMCBIE scheme.

2.3.1. Stage 1: Hénon Chaotic Transform

The Hénon chaos matrix H is generated from the subkeys (x_H, y_H) as described in Section 2.1. The first-stage output $A \in \{0, \dots, 255\}^{M \times N \times 3}$ is obtained by applying a pixel-wise XOR between P and H across all three colour channels:

$$A_{r,c,k} = P_{r,c,k} \oplus H_{r,c}, \quad k \in \{R, G, B\}, \quad (8)$$

where the same chaos value $H_{r,c}$ is XORed independently into each channel. Because H is entirely determined by (x_H, y_H) , the mapping $P \mapsto A$ is a fixed bijection for any given key.

2.3.2. Stage 2: Key Mixing

The key-mixing phase employs a cyclic modular addition chain on the color components of A using the 192-bit subkey MK . The pixel components of each color component are serialized in row-major order. Denoting the flattened pixel index by $j \in \{0, 1, \dots, MN - 1\}$, the output pixel B_j is formed by cross-channel modular addition:

$$B_j^R = \begin{cases} (A_j^G + MK_G[j \bmod 8]) \bmod 256, & \text{if } j < 8, \\ (A_j^G + MK_G[(j - 8) \bmod 8]) \bmod 256, & \text{if } j \geq 8, \end{cases} \quad (9)$$

$$B_j^G = \begin{cases} (A_j^B + MK_B[j \bmod 8]) \bmod 256, & \text{if } j < 8, \\ (A_j^B + MK_B[(j - 8) \bmod 8]) \bmod 256, & \text{if } j \geq 8, \end{cases} \quad (10)$$

$$B_j^B = \begin{cases} (A_j^R + MK_R[j \bmod 8]) \bmod 256, & \text{if } j < 8, \\ (A_j^R + MK_R[(j - 8) \bmod 8]) \bmod 256, & \text{if } j \geq 8. \end{cases} \quad (11)$$

There are three key observations for the cryptanalysis that follows. First, the channel-rotation sequence ($G \rightarrow R$, $B \rightarrow G$, $R \rightarrow B$) is static and publicly known. Second, the offset values have a period of 8 across the image, resulting in at most eight unique per-channel offsets being used. Finally, this stage is completely linear on the plaintext space (modulo 256) and is unaffected by the chaotic state; when MK is known, this stage becomes easily reversible.

2.3.3. Stage 3: 2D-Logistic Chaotic Transform

The 2D-LSCM chaos matrix L is generated from the subkeys (x_L, y_L, R) as described in Section 2.1. The final cipher image C is produced by pixel-wise XOR of B with L across all three colour channels:

$$C_{r,c,k} = B_{r,c,k} \oplus L_{r,c}, \quad k \in \{R, G, B\}. \quad (12)$$

Like for Stage 1, the identical scalar chaos number $L_{r,c}$ is used on all three color channels at each position of the pixel. The overall encryption process can then be described as

$$C_{r,c,k} = [(P_{r,c,\sigma(k)} \oplus H_{r,c}) + MK_k[(r \cdot N + c) \bmod 8]] \bmod 256 \oplus L_{r,c}, \quad (13)$$

where σ denotes the channel-rotation permutation $\sigma(R) = G$, $\sigma(G) = B$, $\sigma(B) = R$ defined by the key-mixing stage.

2.4. Decryption Process

Decryption is precisely the opposite of the encryption procedure, carried out in the opposite sequence of steps. Using the ciphertext C and key $K = (x_H, y_H, MK, x_L, y_L, R)$, the recipient reconstructs H and L in the same manner as during encryption, which is legitimate since both chaotic matrices are uniquely determined by the key.

Inverse Stage 3 (Logistic XOR).. Since XOR is self-inverse, the key-mix output B is recovered by

$$B_{r,c,k} = C_{r,c,k} \oplus L_{r,c}, \quad k \in \{R, G, B\}. \quad (14)$$

Inverse Stage 2 (Key-Mixing).. The operations of channel-rotation and modular addition are then reversed by a modular subtraction operation in conjunction with channel counter-rotation:

$$A_j^G = (B_j^R - MK_G[j \bmod 8]) \bmod 256, \quad (15)$$

$$A_j^B = (B_j^G - MK_B[j \bmod 8]) \bmod 256, \quad (16)$$

$$A_j^R = (B_j^B - MK_R[j \bmod 8]) \bmod 256, \quad (17)$$

with the index offset switching from $j \bmod 8$ to $(j - 8) \bmod 8$ for $j \geq 8$, mirroring Eqs. (9)–(11).

Inverse Stage 1 (Hénon XOR).. The plaintext is finally recovered by

$$P_{r,c,k} = A_{r,c,k} \oplus H_{r,c}, \quad k \in \{R, G, B\}, \quad (18)$$

using the same chaos matrix H as used during the encryption process. Decryption is only possible when precise information regarding the six subkeys is known since any slight change to either (x_H, y_H) or (x_L, y_L, R) yields an entirely new chaos matrix.

3. Cryptanalysis of MMCBIE

We conduct a full CPA attack on the Multiple Map Chaos Based Image Encryption (MMCBIE) algorithm proposed by Jain et al. [32]. Our attack is able to fully recover the plaintext from any target ciphertext regardless of the secret key. The whole process needs only 256 oracle calls for chosen plaintexts and takes $\mathcal{O}(n^2)$ time.

3.1. Notation and Encryption Model

Let P denote an $n \times n$ RGB plaintext image. We index pixels by their flattened row-major position $k \in \{0, \dots, N-1\}$, where $N = n^2$, and write $P_c[k]$ for colour channel $c \in \{R, G, B\}$. All arithmetic is modulo 256 unless stated otherwise; we denote this ring by $\mathbb{Z}_{256}$. The exclusive-OR operation on $\mathbb{Z}_{256}$ is written $\oplus$.

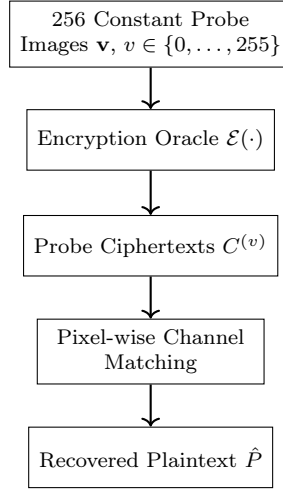


Figure 2: CPA flow for MMCBIE.

MMCBIE applies three successive stages to the plaintext:

Stage 1 — Hénon Chaotic Transform.. A two-dimensional Hénon sequence is generated from subkeys (x_H, y_H) :

$$x_{i+1} = 1 - a x_i^2 + y_i, \quad y_{i+1} = b x_i, \quad (19)$$

with standard parameters $a = 1.4$, $b = 0.3$. The scalar chaos value at pixel position k is

$$H[k] = \lfloor (|x_k| + |y_k|) \cdot 2^{16} \rfloor \bmod 256. \quad (20)$$

The same value $H[k]$ is XORed into every colour channel:

$$P_c^{(1)}[k] = P_c[k] \oplus H[k], \quad c \in \{R, G, B\}. \quad (21)$$

Stage 2 — Key Mixing.. A 192-bit master key is split into three arrays of eight bytes each, $\text{MK}_R, \text{MK}_G, \text{MK}_B \in \mathbb{Z}_{256}^8$. The colour channels are cyclically cross-added with period 8:

$$S_R[k] = (P_G^{(1)}[k] + \text{MK}_G[k \bmod 8]) \bmod 256, \quad (22)$$

$$S_G[k] = (P_B^{(1)}[k] + \text{MK}_B[k \bmod 8]) \bmod 256, \quad (23)$$

$$S_B[k] = (P_R^{(1)}[k] + \text{MK}_R[k \bmod 8]) \bmod 256. \quad (24)$$

The output red channel is a function of the green plaintext channel, the output green channel is a function of the blue plaintext channel, and the output blue channel is a function of the red plaintext channel.

Stage 3 — 2D-Logistic Chaotic Transform.. A two-dimensional logistic sequence is generated from subkeys (x_L, y_L, r) :

$$x_{i+1} = r(3y_i + 1)x_i(1 - x_i), \quad y_{i+1} = r(3x_{i+1} + 1)y_i(1 - y_i), \quad (25)$$

with a 500-iteration warm-up to eliminate transient behaviour. The scalar chaos value at pixel k is

$$L[k] = \lfloor (|x_k| + |y_k|) \cdot 2^{16} \rfloor \bmod 256. \quad (26)$$

The same value $L[k]$ is XORed into every colour channel:

$$C_c[k] = S_c[k] \oplus L[k], \quad c \in \{R, G, B\}. \quad (27)$$

Combining (21)–(27), the complete encryption of the red channel at pixel k with slot index $\ell = k \bmod 8$ is:

$$\boxed{C_R[k] = \left[(P_G[k] \oplus H[k]) + \text{MK}_G[\ell] \right] \bmod 256 \oplus L[k].} \quad (28)$$

Analogous expressions hold for $C_G[k]$ and $C_B[k]$.

3.2. Exploitable Structural Weaknesses

Three independent design flaws make the scheme vulnerable to a CPA.

1. **Plaintext-independent chaos sequences.** The values of $H[k]$ and $L[k]$ are functions of the secret key alone and not of the plaintext. Hence, they can be treated as equivalent-key streams. [34].

2. **Channel-uniform chaos XOR.** The same scalar $H[k]$ (and $L[k]$) transformation is performed on each of the three colour components (refer to (21) and (27)). This reduces three separate degrees of freedom to one, resulting in cross-channel correlation.
3. **Deterministic encryption of constant images.** Since both $H[k]$ and $L[k]$ chaos streams are completely independent of the plaintext, the encryption of all spatially-uniform images, wherein the pixels of a particular channel contain a constant value v , results in a ciphertext whose content is determined solely by v and the key streams. This implies that for every candidate plaintext pixel value $v \in \mathbb{Z}_{256}$, the response $\mathcal{E}(\mathbf{v})$ from the oracle can be used to construct a complete lookup table of all possible encryptions for each pixel. Any hacker who constructs such a table will be able to retrieve, bit by bit, the plaintext pixel value responsible for a ciphertext value without reversing $H[k]$, $L[k]$, or MK.

3.3. Attack Construction

Core Observation.. Fix a pixel position k and consider the red output channel. From (28), for a constant probe image with $P_G[k] = v$ for all k :

$$C_R^{(v)}[k] = \left[(v \oplus H[k]) + \text{MK}_G[k \bmod 8] \right] \bmod 256 \oplus L[k]. \quad (29)$$

As $H[k]$, MK_G , and $L[k]$ are fixed constants that depend only on the key, $C_R^{(v)}[k]$ becomes a deterministic function of v only. Likewise, the green and blue channels in $\mathcal{E}(\mathbf{v})$ become deterministic functions of v only if $P_B[k] = v$ and $P_R[k] = v$, respectively.

Now let C^* be any target ciphertext to be decrypted, with $C_R^*[k]$, $C_G^*[k]$, $C_B^*[k]$ the per-channel values at pixel k . The channel cross-coupling in Stage 2 (equations (22)–(24)) implies:

$$C_R^*[k] \text{ encodes information about } P_G[k], \quad (30)$$

$$C_G^*[k] \text{ encodes information about } P_B[k], \quad (31)$$

$$C_B^*[k] \text{ encodes information about } P_R[k]. \quad (32)$$

This means that, to retrieve the green channel value of the plaintext from pixel k , the attacker must find the value v where the red component value of the probe ciphertext $C^{(v)}$ is equal to the red component of the target ciphertext at position k :

$$\hat{P}_G[k] = v \quad \text{such that} \quad C_R^{(v)}[k] = C_R^*[k]. \quad (33)$$

Analogously:

$$\hat{P}_B[k] = v \quad \text{such that} \quad C_G^{(v)}[k] = C_G^*[k], \quad (34)$$

$$\hat{P}_R[k] = v \quad \text{such that} \quad C_B^{(v)}[k] = C_B^*[k]. \quad (35)$$

Correctness.. For every pixel k and every channel $c \in \{R, G, B\}$, the recovered value satisfies $\hat{P}_c[k] = P_c[k]$.

Consider the green channel recovery at pixel k . Let $v^* = P_G[k]$ be the true plaintext value. Substituting into (29):

$$C_R^{(v^*)}[k] = \left[(P_G[k] \oplus H[k]) + \text{MK}_G[k \bmod 8] \right] \bmod 256 \oplus L[k] = C_R^*[k]. \quad (36)$$

Hence v^* satisfies the matching condition (33), so $\hat{P}_G[k] = P_G[k]$. Identical arguments hold for $\hat{P}_B[k]$ via (34) and $\hat{P}_R[k]$ via (35).

Uniqueness.. The encryption function at each pixel and channel is a composition of XOR with $H[k]$, modular addition of a key byte, and XOR with $L[k]$. This composition is a bijection on $\mathbb{Z}_{256}$ (each operation is individually invertible), so for each pixel k the map $v \mapsto C_R^{(v)}[k]$ is a permutation of $\mathbb{Z}_{256}$. Therefore exactly one value $v \in \{0, \dots, 255\}$ satisfies $C_R^{(v)}[k] = C_R^*[k]$, guaranteeing that the recovered plaintext value is unique.

3.4. Complexity Analysis

Data complexity.. The attack sends precisely 256 plaintext queries to the encryption oracle, one for each value $v \in \{0, \dots, 255\}$ used in probing. The number of queries is independent of image dimensions.

Computational complexity.. Phase 1 requires 256 oracle queries, each encrypting an $n \times n$ image at cost $\mathcal{O}(N)$, giving total query cost $\mathcal{O}(256 \cdot N) = \mathcal{O}(N)$. Phase 2 iterates over N pixels and for each pixel performs at most 256 comparisons across three channels, for a total work of

$$W_{\text{Phase 2}} = N \times 256 \times 3 = 768 N = \mathcal{O}(N). \quad (37)$$

The dominant cost is thus $\mathcal{O}(N) = \mathcal{O}(n^2)$, making the attack comparable in cost to a single encryption, yet achieving complete plaintext recovery without any knowledge of the secret key.

Memory complexity. The attacker stores 256 probe ciphertexts of size $N \times 3$ bytes each, for a total of $\mathcal{O}(256 \cdot N) = \mathcal{O}(N)$ memory.

Algorithm 1 CPA Against MMBCIE

Input: Encryption oracle $\mathcal{E}(\cdot)$; target ciphertext C^* ; image dimensions $n \times n$.

Output: Recovered plaintext $\hat{P}$.

```

// Phase 1 -- Build Probe Lookup Table
1 for  $v = 0$  to 255 do
2   | Construct probe image  $\mathbf{v}$ : set all pixels of all channels to  $v$  Query  $C^{(v)} \leftarrow$ 
   |  $\mathcal{E}(\mathbf{v})$ 
3 end

// Phase 2 -- Pixel-wise Plaintext Recovery via Channel
Matching
4 Initialise recovered image  $\hat{P}$  of size  $n \times n \times 3$  for each pixel position  $k \in$ 
    $\{0, \dots, N - 1\}$  do
5   | for  $v = 0$  to 255 do
6   |   | if  $C_R^{(v)}[k] = C_R^*[k]$  then
7   |   |   |  $\hat{P}_G[k] \leftarrow v$ 
8   |   | end
9   |   | if  $C_G^{(v)}[k] = C_G^*[k]$  then
10  |   |   |  $\hat{P}_B[k] \leftarrow v$ 
11  |   | end
12  |   | if  $C_B^{(v)}[k] = C_B^*[k]$  then
13  |   |   |  $\hat{P}_R[k] \leftarrow v$ 
14  |   | end
15  | end
16 end
17 return  $\hat{P}$ 

```

4. Experimental Simulation of the Chosen-Plaintext Attack

In this part, we provide a full-fledged experimental analysis of the suggested CPA technique in comparison to the MMBCIE protocol developed by Jain et al. [32]. Experiments were carried out using Python 3 and involved

Table 1: Attack complexity versus MNCBIE encryption cost.

	Encryption	CPA Attack
Oracle queries	—	256
Arithmetic operations	$\mathcal{O}(N)$	$\mathcal{O}(N)$
Brute-force search	—	768 N comparisons
Memory	$\mathcal{O}(N)$	$\mathcal{O}(256 \cdot N)$
Key knowledge required	N/A	None

the use of a color image. The CPA conditions were rigorously followed during the experiment such that the attacker had access to the encryption oracle $\mathcal{E}(\cdot)$ but not the secret key or cipher states.

4.1. Experimental Setup

The oracle was configured with the following fixed key, withheld from the attacker:

$$x_H = 0.1, \quad y_H = 0.2, \quad R = 1.15, \quad x_L = 0.3, \quad y_L = 0.4, \quad (38)$$

together with the 192-bit key-mixing subkey $MK = \{MK_R, MK_G, MK_B\}$:

$$\begin{aligned} MK_R &= [12, 34, 56, 78, 90, 11, 22, 33], \\ MK_G &= [44, 55, 66, 77, 88, 99, 10, 21], \\ MK_B &= [31, 42, 53, 64, 75, 86, 97, 18]. \end{aligned}$$

The values served only to create an oracle; at no time during the attack did the attack require information on any of the aforementioned parameters. The cipher image $C = \mathcal{E}(P)$ created using the above parameters does not appear visually similar to the plaintext, as expected for the scheme’s confusion/diffusion features.

4.2. Phase 1: Construction of the Probe Lookup Table

For each candidate plaintext value $v \in \{0, 1, \dots, 255\}$, a spatially uniform probe image $\mathbf{v}$ was constructed in which every pixel of every channel was set to v . Each probe was submitted to the oracle, yielding the ciphertext response $C^{(v)} = \mathcal{E}(\mathbf{v})$. As the Hénon and logistic chaos streams do not depend on the plaintext input at all, the ciphertext output $C^{(v)}$ for each pixel

location k is uniquely determined by v and the key streams alone. Collectively, the 256 images yield a lookup table for each pixel location k and each channel, mapping the 256 different plaintext values to their corresponding ciphertexts. This step precisely utilized 256 oracle queries without needing any information about the secret key.

Table 2: Probe lookup table construction metrics.

Metric	Value
Oracle queries used	256
Probe values covered ($v \in \mathbb{Z}_{256}$)	256
Lookup table fully populated	True
Key knowledge required	None

4.3. Phase 2: Pixel-wise Plaintext Recovery via Channel Matching

For the provided ciphertext C^* and the lookup table created during Phase 1, the attack recovered pixels in a pairwise manner using the cross-coupling property of the second stage channels. As mentioned earlier from the set of equations (22)-(24), the red output channel contains information about the green plaintext channel, green output channel contains information about blue plaintext channel, and blue output channel contains information about red plaintext channel. For every k pixel, the attacker compared all 256 probes with the below matching conditions:

$$\hat{P}_G[k] = v \quad \text{such that} \quad C_R^{(v)}[k] = C_R^*[k], \quad (39)$$

$$\hat{P}_B[k] = v \quad \text{such that} \quad C_G^{(v)}[k] = C_G^*[k], \quad (40)$$

$$\hat{P}_R[k] = v \quad \text{such that} \quad C_B^{(v)}[k] = C_B^*[k]. \quad (41)$$

Because the function for the encryption of each pixel is a bijection from $\mathbb{Z}_{256}$ to itself, exactly one v will fit each match criteria, ensuring uniqueness and correctness at every pixel and channel level. This step involved no more oracle calls, operating in time $\mathcal{O}(256 \cdot N)$.

Table 3: Pixel-wise plaintext recovery metrics.

Metric	Value
Additional oracle queries used	0
Pixel-wise matching conditions applied per pixel	3
Unique match guaranteed per pixel per channel	True
Key knowledge required	None

4.4. Full Decryption Without the Secret Key

The image reconstruction process, $\hat{P}$, consisted of simply arranging the channel matches produced in Phase 2. No part of the real secret key, namely the Hénon parameters, the logistic map parameters, or even any information regarding intermediate cipher states were involved at any stage.

Table 4: CPA decryption accuracy metrics.

Metric	Value
Total oracle queries	256
CPA-decrypted image matches true plaintext ($\hat{P} = P$)	True
Max pixel difference	0
Mean pixel difference	0.000000
Pixel-match rate	100%

Table 5: Summary of the chosen-plaintext attack on the MMCBIE scheme. Total oracle queries: 256.

Phase	Goal	Oracle calls	Items recovered	Perfect match
1	Probe lookup table $\{C^{(v)}\}_{v=0}^{255}$	256	256 full ciphertext images	Yes
2	Plaintext $\hat{P}$ via channel matching	0	Full image, all channels	Yes ($\Delta = 0$)
Total		256		Exact recovery

It is evident from the outcomes that an image encrypted by MMCBIE is possible to recover completely through just 256 queries of the oracle, regardless of the image’s size. Therefore, MMCBIE is *entirely insecure* in the CPA model even considering that it uses a 512 bit-long key. As the following analysis proves, good results for NPCR, UACI, and Entropy do not mean cryptographic security [34].

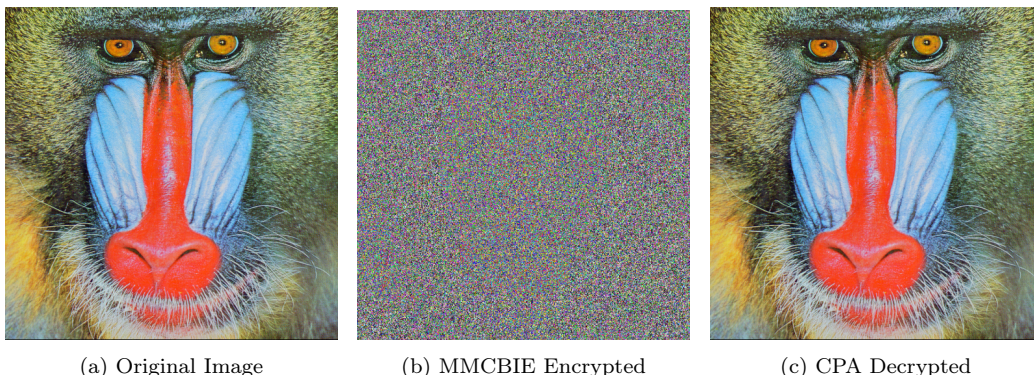


Figure 3: Comparison of original, encrypted, and CPA-decrypted images. The decrypted image is pixel-identical to the original.

The implementation used for all experiments is available at Colab Notebook.

4.5. Root Cause Analysis: Why the Scheme is Broken

Three mutually reinforcing design defects collectively explain the attack’s success.

Defect 1: Plaintext-independent chaotic keystreams. However, since both the Hénon map and the 2D-Logistic map have their initial seeds drawn solely from their corresponding fixed subkeys (x_H, y_H) and (x_L, y_L, R) , without taking any inputs from the plaintext image, the chaos matrices that will be produced by them will remain consistent throughout the entire set of images encrypted using the exact same key. In fact, a cipher where its keystream does not depend on the plaintext itself will essentially become a one-time pad cipher, which is inherently insecure to attacks involving chosen-plaintext [34]. This is precisely why the probe lookup table works as expected: encryption of the same constant value v will result in the same ciphertext, no matter what the plaintext actually is.

Defect 2: Bijectivity of the per-pixel encryption function. The application of the XOR operation on $H[k]$, the addition of the key byte mod 256, and the XOR operation on $L[k]$ yields a bijection in $\mathbb{Z}_{256}$. Although the property of bijectivity can be viewed positively in block ciphers, in our context it ensures that the 256-probe ciphertexts in the lookup table will contain all possible values for any pixel position k and any particular channel, without any duplications whatsoever. Hence the matching criteria in equa-

tions (39)–(41) will result in an unambiguous determination of the plaintext, ensuring accuracy.

Defect 3: Channel cross-coupling without plaintext dependence.

During the Key Mixing phase, the green plaintext channel feeds into the red ciphertext channel, the blue plaintext channel feeds into the green ciphertext channel, while the red plaintext channel feeds into the blue ciphertext channel. It appears that the cross-coupling was added for the purpose of introducing confusion between the channels, but since the chaotic streams are independent of the plaintext, they can easily be removed by means of the probe-and-match approach. With one constant probe, all three channels will have the same value v , thus only 256 oracle queries are necessary to break all three plaintext channels in parallel. In case the chaotic streams depended on the plaintext, the cross-coupling would introduce some nonlinearities.

When all the weaknesses occur at once, the protection offered by the proposed scheme becomes equivalent to the process of table look-up only. The adversary will not have to find out the secret key, reverse any chaotic function, or solve any set of equations. The whole idea will boil down to simply encrypting 256 images and then looking up for the ciphertext value in a pre-built lookup table in order to get the plaintext values. This makes the proposed scheme insecure in the chosen-plaintext setting.

4.6. Recommendations for Strengthening Against Chosen-Plaintext Attacks

The analysis in this paper highlights three essential flaws within the structure of MMCBIE, which together facilitate an elementary chosen plaintext attack. The following suggestions aim to rectify each flaw individually, providing insight into the construction of future chaos-based image encryption techniques.

4.6.1. Plaintext-Dependent Chaos Initialisation

The single most important change, however, would be to integrate the chaos in the generation of the keystream directly into the plaintext itself. As explained in Section 4.5, since neither $H[k]$ nor $L[k]$ depends on the plaintext at all, MMCBIE boils down to using a reusable pad, which is unconditionally insecure against chosen plaintext attacks.

- **IV perturbation using hashes.** Before applying encryption, calculate the hash of the plaintext image $\mathcal{H}(P)$ and combine it via an XOR operation with the chaotic initial states for both generators. As the

value of $\mathcal{H}(P)$ varies for each new plaintext, each pair of chaos matrices H and L becomes unique for each new plaintext, rendering the probe LUT useless when applied to a different plaintext.

- **Feedback coupling in chaos generation.** Instead of generating the chaos from static initial conditions, use the encrypted pixels periodically to reset the chaotic system state through some function. This results in dependencies between image regions, meaning that encrypting a probe image consisting of identical pixel values will no longer replicate the oracle behaviour on a real image.
- **Adding nonce into IV.** In addition to the chaotic initial conditions, a unique random value can be used for each encryption and included into the IV, ensuring that even two identical plaintexts result in different ciphertexts and hence cannot be probed successfully.

4.6.2. Strong Inter-Pixel Diffusion

The separability of the MMCBIE encryption algorithm in terms of pixels is what makes it possible for the attack to run separately for each pixel location. It is necessary for a robust technique to be able to make changes spread throughout the entire area of the image such that the encryption of any one pixel is dependent on other pixels.

- **Global permutation before the diffusion step.** Performing a global pixel permutation—dependent on the plaintext—such as the Arnold cat map based on the hash of the plaintext itself—before the diffusion stages renders impossible the assumption of independence upon which the CPA relies. The permutation of the pixels will ensure that the ciphertext of the probe will not be identical to that of the plaintext, since Equations (39) and (41) will not hold anymore.
- **Chaining diffusion with carry propagation.** Chaining together the diffusions in Stage 3 through a sum modulo 256, for instance,

$$C[k] = (B[k] \oplus L[k] \oplus C[k - 1]) \bmod 256, \quad (42)$$

will guarantee that any given ciphertext value is dependent upon the entire sequence before. In turn, changing one bit of the plaintext will cause an avalanche effect in the entire ciphertext, overcoming the lack of this effect in the scheme.

- **Two or more diffusion rounds.** By performing two or more rounds of permutation and diffusion, the dependency of each ciphertext value on a greater number of plaintext values renders the independence assumption for the attack invalid.

4.6.3. Plaintext-Dependent and Non-Linear Key Mixing

However, in the key mixing stage of MMCBIE, the linear modular addition is deterministic, using a key stream with a period of 8. Hence, due to this deterministic approach, there is no security provided by this stage once the chaos streams become vulnerable to the probe table, because there are just eight points at which the exploit could be done.

- **Use of substitution box.** The substitution of the modular addition with a substitution box (S-box), based on e.g., a chaotic Latin square [18], adds true nonlinearity and stops the mapping from being a bijective function whose enumerations take only 256 oracle queries.
- **Key schedule with period cycling.** Expanding the lengths of key-mixing arrays MK_R , MK_G , and MK_B to MN from 8 removes the periodicity that limits the complexity of Stage 2 to 8 bits. In actual implementation, this can be done by generating a longer subkey with a pseudo-random generator seeded with the master key.
- **Data-driven cross-channel operations.** As an improvement over a predetermined rotation pattern (G to R, B to G, R to B), the cross-channel relationships must be determined using either the plaintext data or a function of the key/plaintext combination having high entropy.

4.6.4. Security Validation Beyond Statistical Metrics

A general lesson from this study is that good scores for NPCR, UACI, and Shannon entropy cannot guarantee cryptographic security [34]. These metrics provide information about the statistical behavior of the ciphertext distribution but fail to evaluate resistance against active attacks in the chosen plaintext/ciphertext model scenarios. For future designs, security guarantees such as IND-CPA security need to be verified along with applying standard cryptanalytic methods—differential, linear, and algebraic attacks. Consequently, this will prevent the use of flawed schemes in security-sensitive IoT settings.

5. Conclusion

For the analysis, we use the chosen-plaintext model to examine the chaotic image encryption technique MMCBIE presented by Jain et al.[32]. Despite being based on several chaotic maps, including the Hénon map, a cyclic key mixing step, and a two-dimensional logistic map, the technique has been proven to be vulnerable because of weak diffusion and inherent structural properties.

Weaknesses in the technique arise from insufficient dependency of adjacent pixels and channels. Transformations applied in the encryption technique are performed independently and follow a certain pattern; thus, there still remain correlations between corresponding plaintext and ciphertext values. Specifically, dependencies arising in the key mixing operation applied on each channel remain unchanged; moreover, the lack of feedback leads to ineffective diffusion of alterations.

Based on the above features, an efficient attack strategy is developed. By making requests to the oracle encryption service for constant images within the whole range of pixels, we obtain cipher texts. Comparing them with the target ciphertext using simple equality checks allows us to retrieve original plaintext values bit by bit. It should be noted that in this way, we do not need to restore chaotic series, secret parameters, or keys.

Theoretically and experimentally, it was proved that using the above attack, exact restoration of the original image is possible using no more than 256 queries. In other words, we demonstrate a total failure of the encryption scheme despite its excellent results in statistics tests such as NPCR, UACI, and entropy.

These findings confirm one of the basic tenets of cryptography: Statistical randomness is not enough to ensure security from active attacks. In order for chaos-based image cryptosystems to be considered secure, it is imperative that they use diffusive functions, have inter-pixel and inter-channel interactions, and employ non-linear functions that depend on the plain-image input.

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